# Evidence and Sources

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Transparent Book Recommender

THE CONNECTED WEEK

## One connected claim-support record.

Explain the rubric, trace the counts, audit their limit and plan a source check.

**Keep one habit:** Retain claim and citation denominators, source passages, corrections and unresolved facts.

MAIN LESSON · SECTION 1

## Recall: reconnect earlier ideas

> Retrieve before opening notes.

### Retrieve and explain

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from Prompt Experiments. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Image and Voice Pipelines. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Recommendation Scores. Give an example and a mistake that the idea helps you avoid.

**My remembered idea, example and correction**

**My remembered idea, example and correction**

**My remembered idea, example and correction**

▼ Earlier wording, alternatives and picture transcripts

**Prompt Experiments**

**Image and Voice Pipelines**

**Recommendation Scores**

## W1 · Explain exact claim support

> Name the claim, cited passage and inference boundary.

### Your evidence record

- Four-claim W2 audit: C1,C2 have real citations that support them; C3 has a real but non-supporting citation; C4 has no citation. Support is by cited passage. Separate Ocean Maps picture has three claims, all linked to R, but only page count and shelf supported. R says80pages,shelf2 and gives no suitability assessment. These are different answers and denominators.

- Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

**My reasoning and evidence**

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MAIN LESSON · SECTION 13

## W2 · Trace four claims

> Compute separate support, citation-presence and link-resolution rates.

### Your evidence record

- Four-claim W2 audit: C1,C2 have real citations that support them; C3 has a real but non-supporting citation; C4 has no citation. Support is by cited passage. Separate Ocean Maps picture has three claims, all linked to R, but only page count and shelf supported. R says80pages,shelf2 and gives no suitability assessment. These are different answers and denominators.

- An answer has four factual **claim**s. Three have citations, but only two are supported by their cited passages. All links exist. Show the intermediate steps and give the conclusion with its scope.

> **My reasoning and evidence**

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MAIN LESSON · SECTION 14

## W3 · Audit what the counts cannot establish

> Separate cited support, missing evidence and real-world truth.

### Your evidence record

- Four-claim W2 audit: C1,C2 have real citations that support them; C3 has a real but non-supporting citation; C4 has no citation. Support is by cited passage. Separate Ocean Maps picture has three claims, all linked to R, but only page count and shelf supported. R says80pages,shelf2 and gives no suitability assessment. These are different answers and denominators.

- A report omits this qualification: Citation counts do not establish truth beyond the supplied **source**s or the correctness of uncited **claim**s.

  Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

> **My reasoning and evidence**

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MAIN LESSON · SECTION 16

# W5 · Design a revealing evidence check

> Preserve exact editions, passages and source identities.

## Your evidence record

- Four-claim W2 audit: C1,C2 have real citations that support them; C3 has a real but non-supporting citation; C4 has no citation. Support is by cited passage. Separate Ocean Maps picture has three claims, all linked to R, but only page count and shelf supported. R says80pages,shelf2 and gives no suitability assessment. These are different answers and denominators.

- For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

  Do not reuse W4 as a new final test after receiving feedback.

**My reasoning and evidence**

MAIN LESSON · SECTION 4

# W4 · Audit a new claim-and-citation record independently

> Keep one first attempt before hints or feedback.

## Read and explain independently

For this independent counting task, treat each cited claim as having one citation and the nonexistent ID as the only unresolved link; the other three citations resolve. The supplied support count is by cited passage, not an assertion that uncited claims are false. No additional passage content is supplied, so do not invent it.

Five **claim**s; four cited; three supported by cited passages; one cited ID is nonexistent. Compute support and citation presence and explain link limits.

**My independent explanation, trace and limits**

## Keep a fair record

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

▼ Earlier wording, alternatives and picture transcripts

**Independent case: Apply a new case independently**

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MAIN LESSON · SECTION 9

## Project · Keep the decision and its evidence

> Add this week's change to your existing project record.

### Your evidence record

Separate profile-based ranking **evidence** from factual book information and cite the latter.

The project change and **evidence** I will keep

> **Write or draw your reasoning and evidence.**

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MAIN LESSON · SECTION 17

## Feedback · Repair and return

> Keep the first attempt, the repair and a changed-case return.

### Your evidence record

| Evidence | My record |
|---|---|
| Worked alone / hint / model | |
| First step I repaired and why | |
| New case checked later; date and result | |
| What I can now explain without notes | |
| One remaining question | |

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

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MAIN LESSON · SECTION 10

## Finish · Judge the evidence in your work

> Point to work that supports your confidence level.

### Your evidence record

- I can explain the mechanism.
- I can show evidence.
- I can handle a changed case.
- I can name a limit.

Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

# Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

## Reference · Supported W2 record

> This is the same W2 attempt. Keep one response record.

### Use this when helpful

Use W2: Trace the worked case for **Claim** support and citation evaluation. Record the answer once; the lesson and workbook refer to the same attempt.

| Given inputs | Procedure I chose | Middle steps | Result and check |
|---|---|---|---|
| | | | |

> **The step I completed without help, and the hint I needed**

▼ Extra practice · Repair the explanation

REFERENCE · SECTION 3

## Extra practice · Repair the explanation

> Explain the deciding fact before discussing the answer.

### Use this when helpful

Use the worked case for **Claim** support and citation evaluation. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Original learning target

REFERENCE · SECTION 5

## Reference · Original learning target

> Use this target to inspect an explanation.

## Use this when helpful

Two catalogue records list 96 and 112 pages for a title. A helper averages them and reports 104. Explain the mistake and a better verification **process**.

▼ **Extra practice · A complete case**

REFERENCE · SECTION 6

# Extra practice · A complete case

> Try the question before checking the worked explanation.

## Use this when helpful

An answer says 80 pages, but the catalogue for that edition says 120.

What **evidence** guides correction?

Show your trace, calculation or **evidence**

> **Write or draw your reasoning and evidence.**

▼ **Extra practice · A changed case**

REFERENCE · SECTION 7

## Extra practice · A changed case

> Keep this practice separate from the reserved independent attempt.

### Use this when helpful

An answer cites a page about opening hours to support a **claim** about book prices. Is a citation enough? What must be checked?

My practice answer and reason

> **Write or draw your reasoning and evidence.**

**Week 29 Test and reflect**

▼ **Extra practice · Design a revealing test**

REFERENCE · SECTION 8

## Extra practice · Design a revealing test

> Predict before tracing or testing.

### Use this when helpful

Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

| Test record | My evidence |
| --- | --- |
| Changed input or condition | |
| Expected answer or fallback | |
| Observed result | |
| Why the result occurred | |

▼ **Reference · Original target response space**

REFERENCE · SECTION 11

# Reference · Original target response space

> This teaching case is practice rather than a fresh assessment.

## Use this when helpful

Two catalogue records list 96 and 112 pages for a title. A helper averages them and reports 104. Explain the mistake and a better verification **process**.

**My explanation after practice**

Record support used: none, hint or teacher explanation.

▼ Reference · Independent-case response copy

REFERENCE · SECTION 15

# Reference · Independent-case response copy

> This duplicates W4. Reuse the same first-attempt record.

## Use this when helpful

Five **claim**s; four cited; three supported by cited passages; one cited ID is nonexistent.
Compute support and citation presence and explain link limits.

**My reasoning and evidence**

▼ Extra practice · Use the worked picture

REFERENCE · SECTION 18

# Extra practice · Use the worked picture

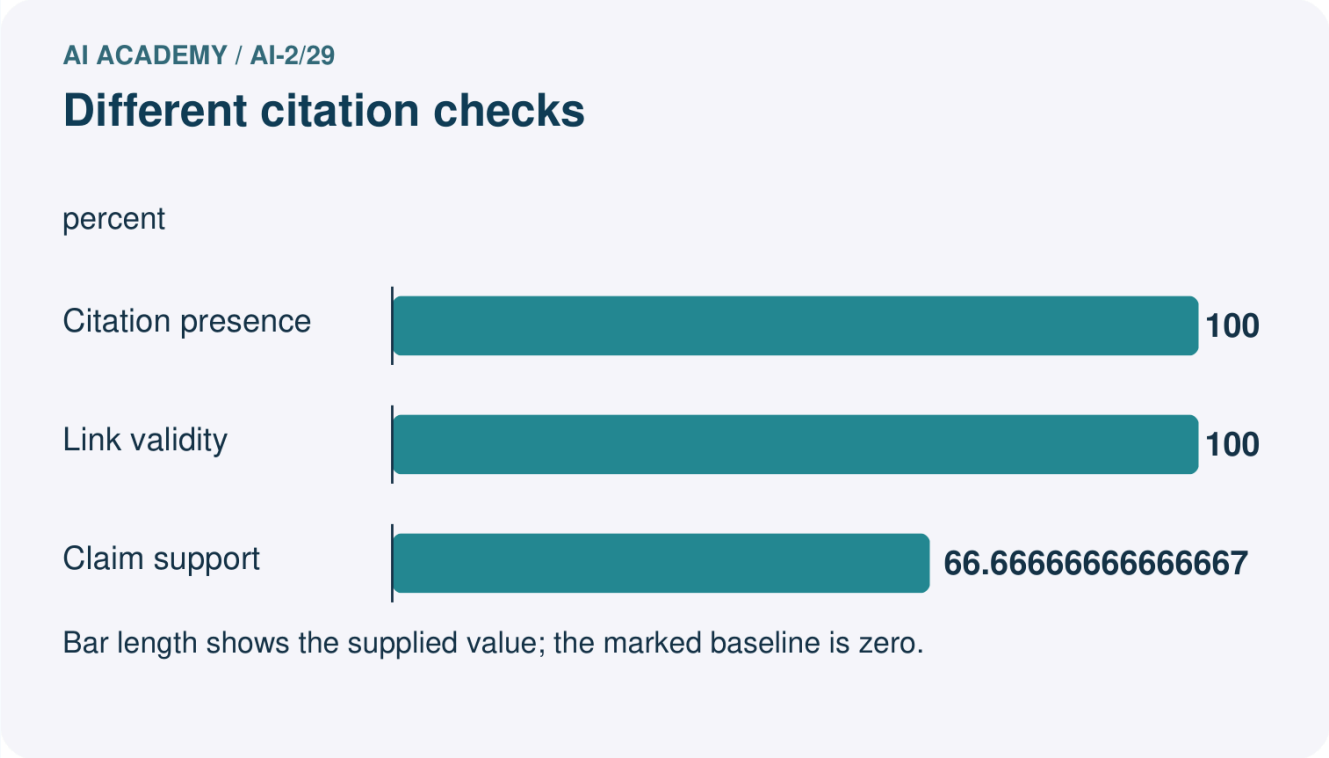
> Keep the explanation and the support you used.

## Use this when helpful

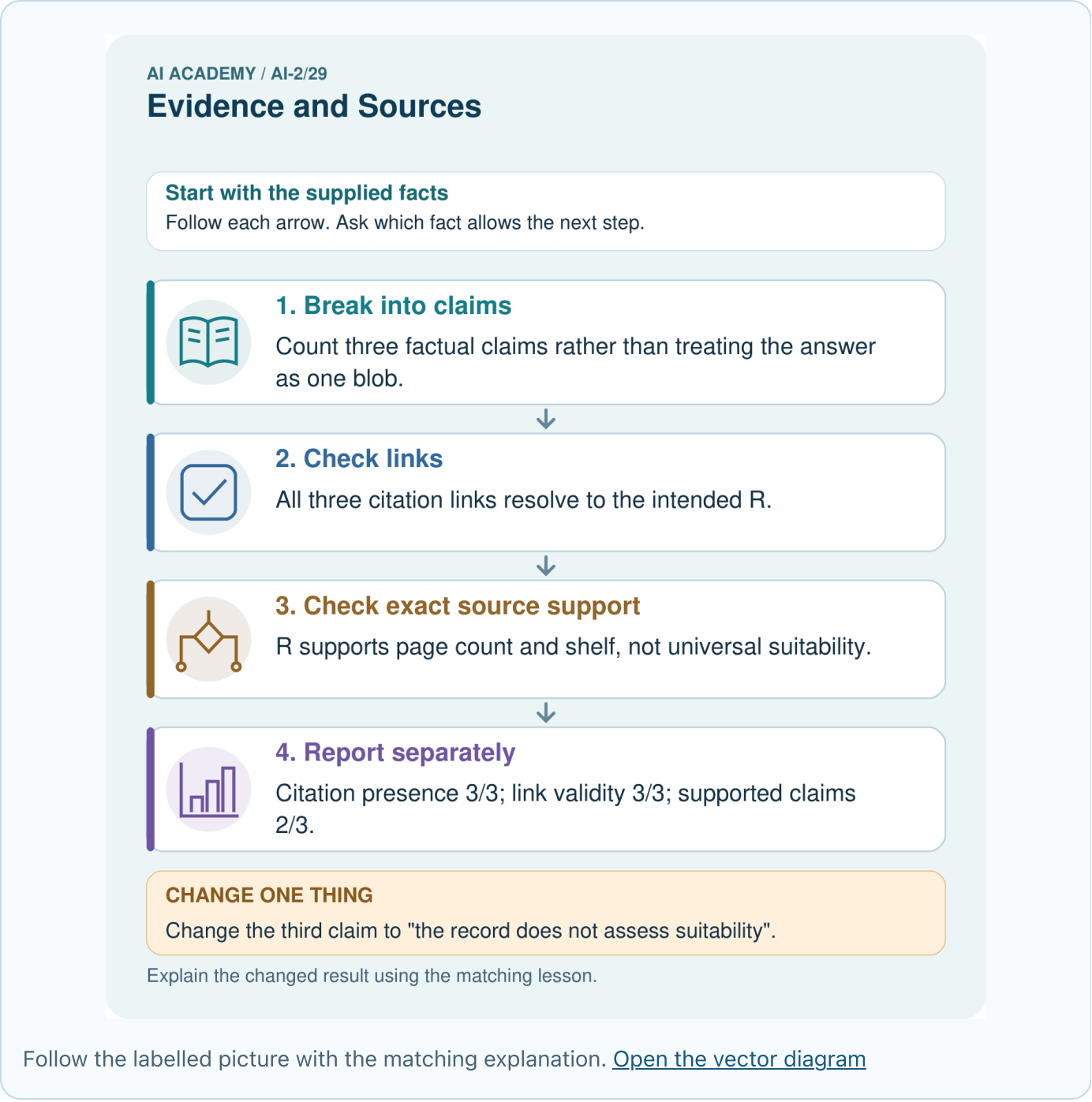
A fictional catalogue answer contains three factual statements and cites existing records.

Record R: book Ocean Maps has 80 pages and is available on shelf 2.

**Claim**s: 80 pages [R]; shelf 2 [R]; suitable for every child [R]. All three citations resolve to R.




Follow the labelled picture with the matching explanation. Open the vector diagram




Follow the labelled picture with the matching explanation. Open the vector diagram

Why is the original answer not fully supported despite having citations on every **claim**?

### My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: Attach **source** IDs to recommender explanation facts and separately audit whether each **source** supports its precise statement.

▼ **Reference · Project portfolio record**

REFERENCE · SECTION 19

# Reference · Project portfolio record

> Extend your main project response when helpful.

## Use this when helpful

Project: Transparent Book Recommender

Current milestone: Prepare the demonstration. Gather reproducible instructions, **evidence**, access checks and remaining limitations.

Attach **source** IDs to recommender explanation facts and separately audit whether each **source** supports its precise statement.

| Evidence to keep | My record |
|---|---|
| This week's input and deciding step | |
| Prediction and actual result (or labelled paper trace) | |
| One change since the previous checkpoint and why | |
| One error or limit and the next test | |

> **Explain what this week adds to your project. Reuse last week's record rather than starting again.**

▼ **Extra practice: pictures, repair and feedback**

REFERENCE · SECTION 20

# Extra practice: pictures, repair and feedback

> Attempt each question before opening its explanation. Use one browser or paper copy.

## First supported question

- A fictional catalogue answer contains three factual statements and cites existing records.
- Which labelled step matches this detail? Citation presence 3/3; **link validity** 3/3; supported **claim**s 2/3.
- 1. Check exact **source** support
- 2. Check links
- 3. Report separately
- 4. Break into **claim**s
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

### Practice 1: explain the deciding evidence

**My selected conclusion, deciding fact and explanation**

▼ **Check the first explanation**

After your attempt, compare with the supported explanation: Report separately Citation presence 3/3; **link validity** 3/3; supported **claim**s 2/3. An existing citation is not automatically **evidence** for every nearby sentence.

Read the cited passage and compare its exact scope with the **claim**. The universal suitability statement extends beyond what R says.

Remove or qualify unsupported content; adding more copies of the same link does not create the missing **evidence**.

## Changed-case question

- Changed-case practice: Change the third **claim** to “the record does not assess suitability”. Which conclusion is supported by the supplied facts?
- 1. That statement accurately describes the supplied record’s limits.
- 2. This one example guarantees correct results for every future **input**.
- 3. We can decide the result without examining the changed **input** or the procedure.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

### Practice 2: explain the deciding evidence

**My selected conclusion, deciding fact and explanation**

▼ **Check the changed case**

After your attempt, compare with the supported explanation: That statement accurately describes the supplied record’s limits. R gives pages and shelf only; it contains no suitability assessment.

## Repair and save your work

Review Prompt Experiments (ai-2/28). Goal: Explain the **mechanism** and terminology of controlled prompt experiments. Short practice: Name the case P passes and Q fails, and explain why it matters..

Return to this week and explain what you changed.

Review Image and Voice Pipelines (ai-2/26). Goal: Explain the **mechanism** and terminology of speech pipelines and multimodal **evidence**.

Short practice: Which stage first fails, and why should the faulty transcript not be silently overwritten?. Return to this week and explain what you changed.

Review Recommendation Scores (ai-2/23). Goal: Compute a weighted score. Short practice: Why does B not become the winner when its score is largest?. Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

**What I repaired, the evidence I kept and the next test**

▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved **independent** assessment for a separate first attempt before teacher feedback.

**Repair a missing building block**

**Keep your completed work**

▼ **Reference · Six-picture response sheets**

REFERENCE · SECTION 21

## Reference · Six-picture response sheets

> Choose the image that helps rather than repeating all six.

# 01. Evidence and Sources: the case

AI ACADEMY / AI-2/29 · 01 · LOOK AT THE FACTS

## Evidence and Sources: the case

### The situation

A fictional catalogue answer contains three factual statements and cites existing records.

### Supplied evidence

Record R: book Ocean Maps has 80 pages and is available on shelf 2.
Claims: 80 pages [R]; shelf 2 [R]; suitable for every child [R]. All three citations resolve to R.

**By the end, I can separate citation presence, link validity and support for an exact claim.**

Follow the labelled picture with the matching explanation. Open the vector diagram

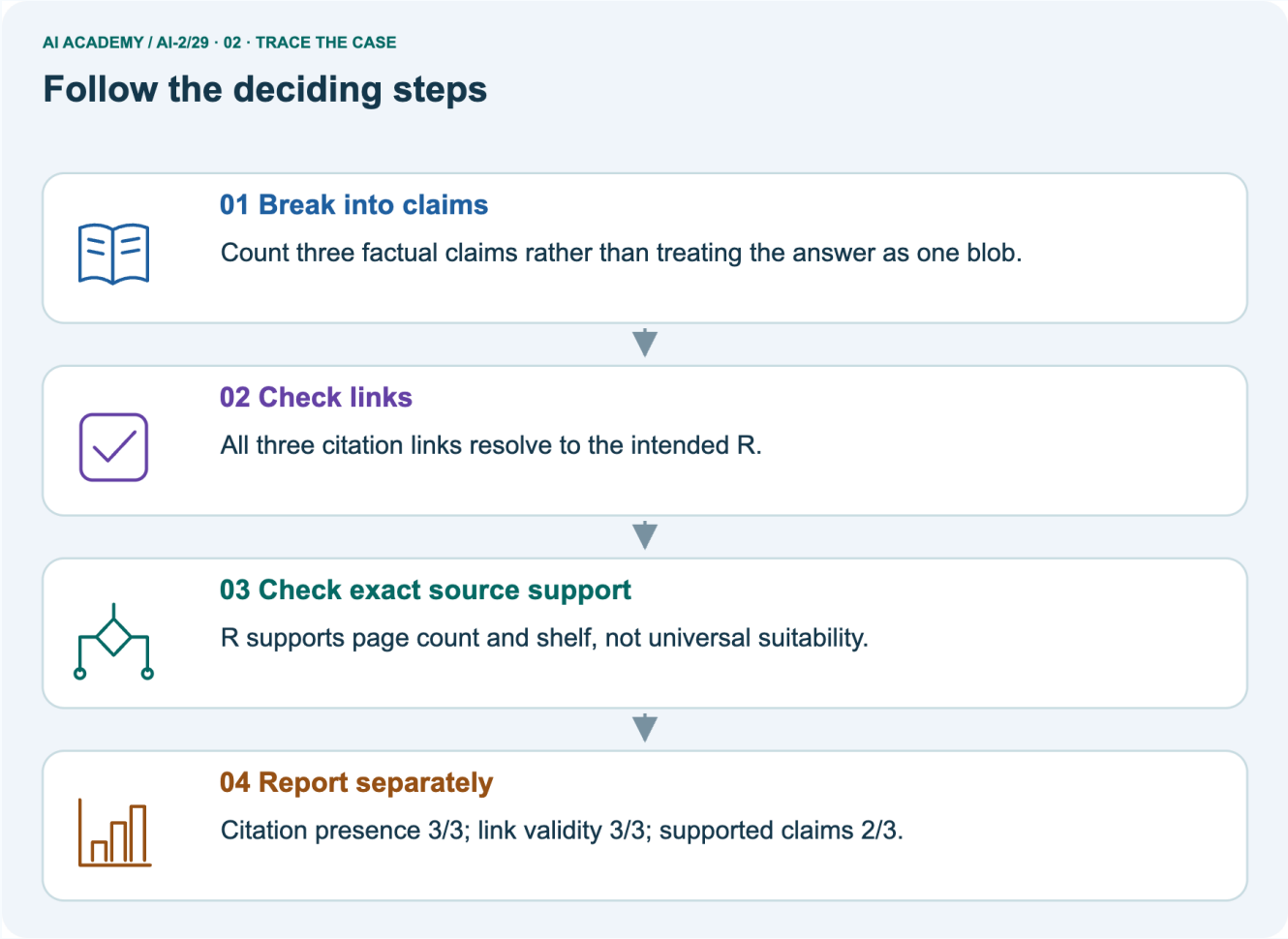
A fictional catalogue answer contains three factual statements and cites existing records.
Supplied facts: Record R: book Ocean Maps has 80 pages and is available on shelf 2.

**Claim**s: 80 pages [R]; shelf 2 [R]; suitable for every child [R]. All three citations resolve to R.

Point to the given fact that matters most. Explain why it matters.

### My picture explanation, deciding evidence and one limit

## 02. Follow the deciding steps



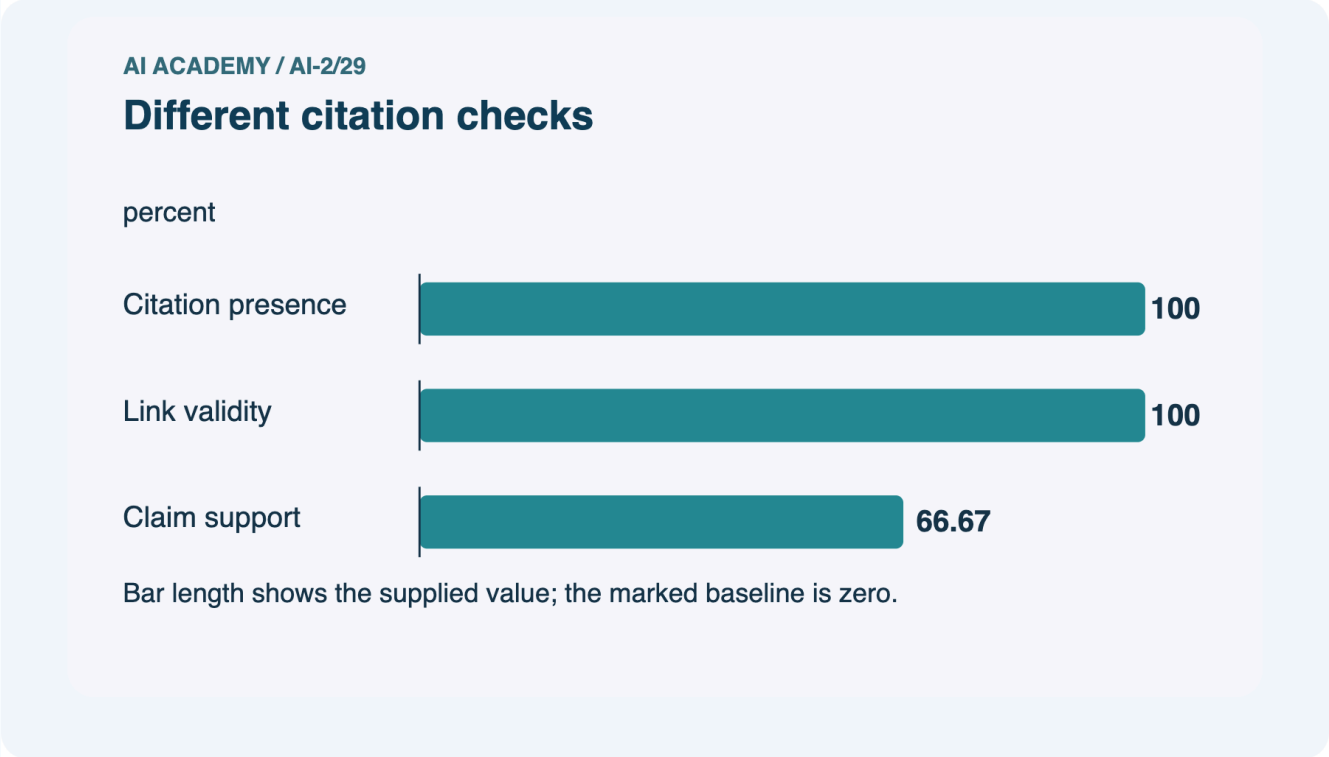

Follow the labelled picture with the matching explanation. Open the vector diagram

Break into **claim**s: Count three factual **claim**s rather than treating the answer as one blob. → Check links: All three citation links resolve to the intended R. → Check exact **source** support: R supports page count and shelf, not universal suitability. → Report separately: Citation presence 3/3; **link validity** 3/3; supported **claim**s 2/3.

Point to each numbered stage and explain which recorded **input** it uses.

### My picture explanation, deciding evidence and one limit

## 03. Different citation checks



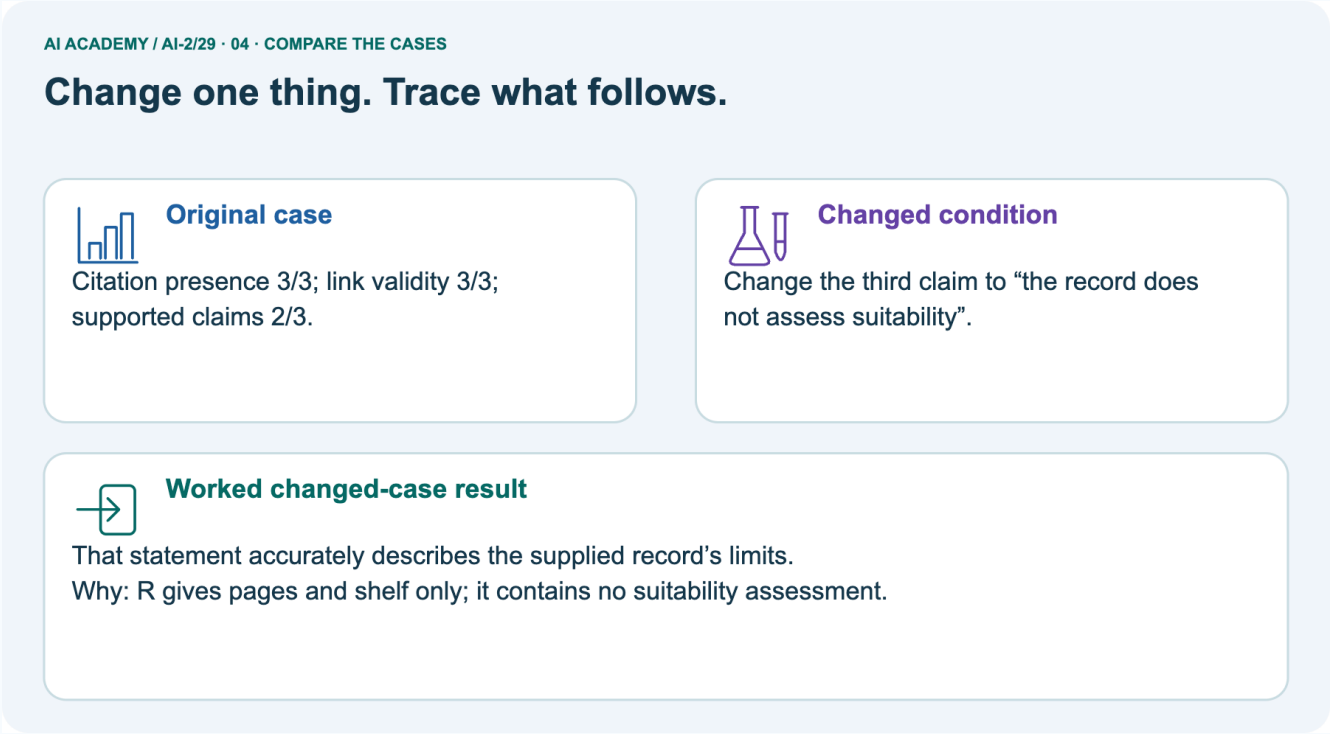

Follow the labelled picture with the matching explanation. Open the vector diagram

Different citation checks. Units: percent. Citation presence: 100. **Link validity**: 100. **Claim** support: 66.66666666666667.

Read the labels and values. Explain the relationship the model shows.

### My picture explanation, deciding evidence and one limit

## 04. Change one thing. Trace what follows.




Follow the labelled picture with the matching explanation. Open the vector diagram

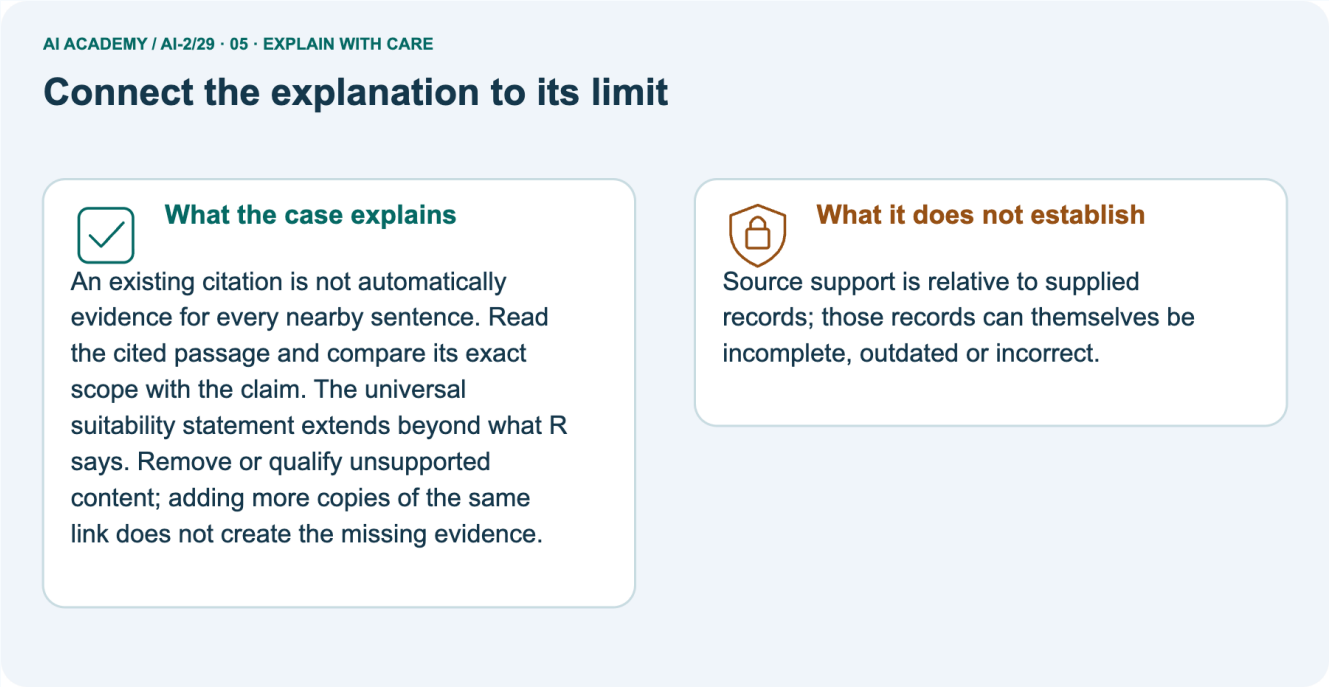
Original case: Citation presence 3/3; **link validity** 3/3; supported **claim**s 2/3. Changed condition: Change the third **claim** to “the record does not assess suitability”.

Expected result: That statement accurately describes the supplied record’s limits. Reason: R gives pages and shelf only; it contains no suitability assessment.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

### My picture explanation, deciding evidence and one limit

## 05. Connect the explanation to its limit




Follow the labelled picture with the matching explanation. Open the vector diagram

An existing citation is not automatically **evidence** for every nearby sentence. Read the cited passage and compare its exact scope with the **claim**. The universal suitability statement extends beyond what R says.

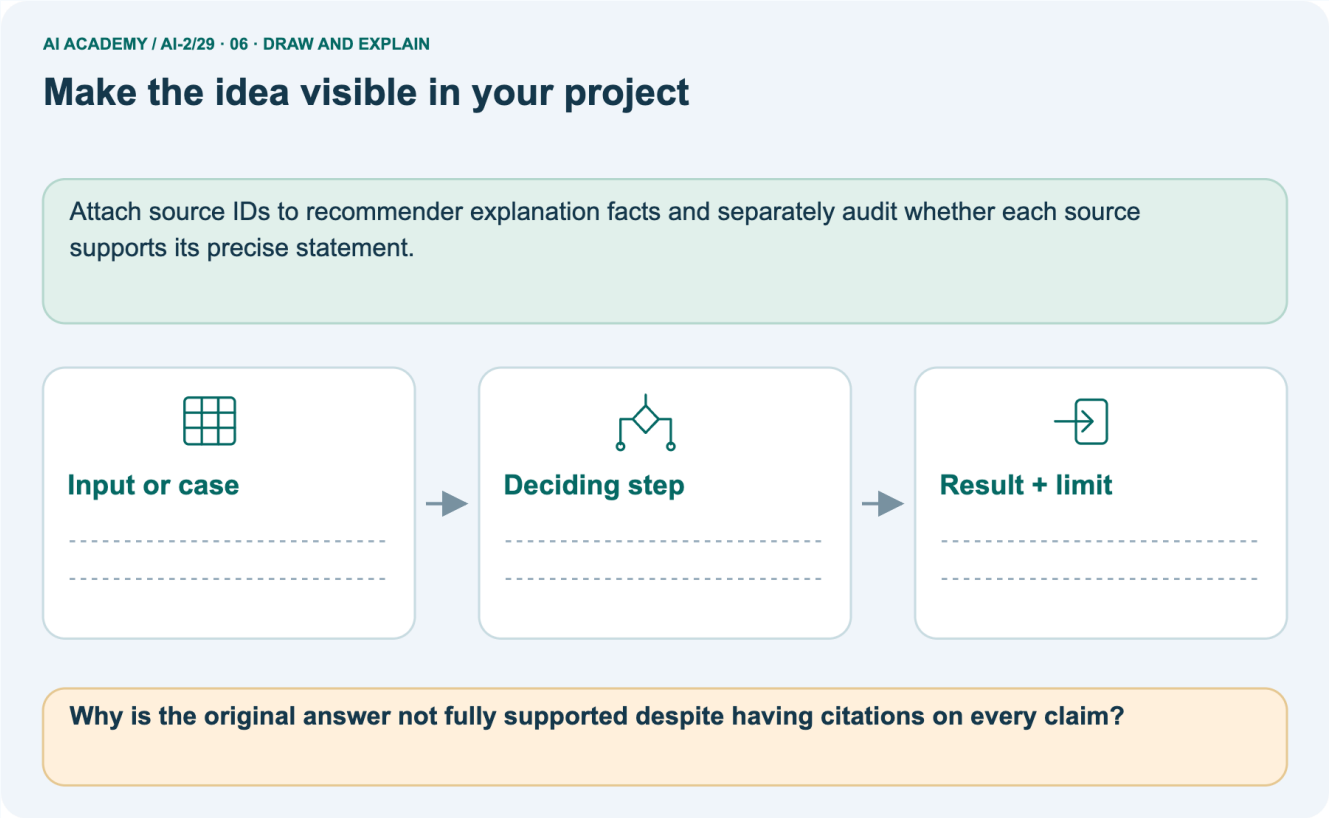
Remove or qualify unsupported content; adding more copies of the same link does not create the missing **evidence**.

Limit: **Source** support is relative to supplied records; those records can themselves be incomplete, outdated or incorrect.

Explain the case, then point to a limit on the **claim**.

### My picture explanation, deciding evidence and one limit

## 06. Make the idea visible in your project




Follow the labelled picture with the matching explanation. Open the vector diagram

Attach **source** IDs to recommender explanation facts and separately audit whether each **source** supports its precise statement. Draw your **input**, deciding step, result and limit.

Supported question: Why is the original answer not fully supported despite having citations on every **claim**?

Draw your project trace and justify the deciding step using **evidence**.

### My picture explanation, deciding evidence and one limit

## ▼ Earlier wording, alternatives and picture transcripts

### 01. Evidence and Sources: the case

### 02. Follow the deciding steps

### 03. Different citation checks

### 04. Change one thing. Trace what follows.

### 05. Connect the explanation to its limit

### 06. Make the idea visible in your project

## ▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.